CIVIL ENGINEERING					
IV B.Tech – I Semester					
Course Code	**SUBSURFACE INVESTIGATION AND INSTRUMENTATION Professional Elective-V**	L	T	P	C
23A01703b		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamental concepts of soil formation, classification, and stratification processes.
2. **Analyze** various soil exploration methods, including boring, drilling, and sampling techniques.
3. **Evaluate** borehole logging methods and groundwater observations for site investigation.
4. **Apply** field testing techniques such as SPT, PLT, PMT, CPT, and geophysical methods to assess soil properties.
5. **Assess** soil exploration report preparation, including instrumentation and data interpretation.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. **Explain** soil formation processes, classification methods, and stratification phenomena.
2. **Analyze** soil exploration methods and sampling techniques for various geotechnical applications.
3. **Evaluate** borehole logging, groundwater observations, and their influence on soil properties.
4. **Apply** field testing procedures, including penetration tests and geophysical methods, to assess subsurface conditions.
5. **Assess** soil exploration report writing and field instrumentation techniques for site investigations.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	1	2	1	2	2	1	-	2	-	2	2	2
CO -2	3	3	2	3	2	2	3	1	-	2	-	2	2	2
CO -3	3	3	3	3	3	2	3	2	-	2	-	2	3	3
CO -4	3	2	3	3	3	3	3	2	-	2	2	2	3	3
CO -5	3	3	3	3	3	3	3	2	-	2	2	2	3	3

UNIT – I

Introduction: Soil formation, Types of Soils, Physical and Chemical Weathering, Soil Transport, Deposition and Stratification Phenomena and Soil Classification.

UNIT – II

Methods of Soil Exploration: Methods of Boring, Auguring and Drilling. Machinery Used for

Drilling, Types of Augers and Their Usage for Various Projects. Soil Sampling: Sampling Methods, Types of Samples, Storage of Samples and Their Transport. Sample Preparation, Sample Sizes, Types of Sampler's Specifications for Testing.

UNIT – III

Borehole Logging: Logging of Boreholes - Logging Methods - Groundwater Observations – Water Table Fluctuations and Effects - Preparation of Soil Profiles and Exploration Report.

UNIT – IV

Field Testing of Soils: Methods and Specifications – Visual Identification Tests, Standard Penetration Test (SPT), Plate Load Test (PLT), Pressure Meter Test (PMT) Dilatometer Test (DMT) Vane Shear Test (VST), Cone Penetration Test (CPT), Becker Penetration Test (BPT), Analysis of Test Results. Geophysical Methods of Soil Exploration- Seismic Refraction, Electrical Resistivity, Cross Hole Test.

UNIT – V

Report Writing: Soil Exploration Reports- Identification, Calculations and Preparation. Field Instrumentation: Strain Gauges, Piezometer, Pressure Cells, Inclinometers, Proving Ring, Load Cells, Displacement Gauges

TEXT BOOKS:

1. Site Investigation, Clayton C. R., Matthews M. C and Simons N. E., Blackwell Science. 2005
2. Geotechnical Instrumentation for Monitoring Field Performance, John Dunn cliff, WileyInterscience, 2008

REFRENCE BOOKS:

1. Basic and Applied Soil Mechanics- A.S. Rao andGopalRanjan, New Age International.
2. IS:1892-Code of Practice for subsurface investigation for foundation, 1979
3. IS: SP36 Part 1-Compendium of India Standards on Soil Engineering-Laboratory Testing of Soils for Civil Engineering Purposes, 1987.
4. IS: SP36 Part 2-Compendium of India Standards on Soil Engineering-Field Testing of Soils for Civil Engineering Purposes, 1988.

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/103/105103182/
https://onlinecourses.nptel.ac.in/noc25_ce27/preview
https://onlinecourses.nptel.ac.in/noc22_ce81/preview